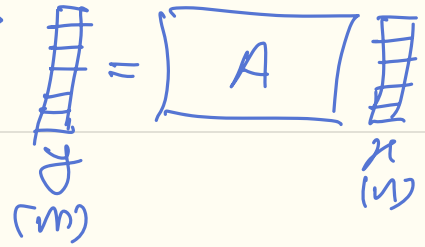


$$y = Ax \quad y \in \mathbb{R}^m, x \in \mathbb{R}^n \rightarrow$$

① $m = n, A = \text{sq.}$

$$\hat{x} = A^{-1}y$$



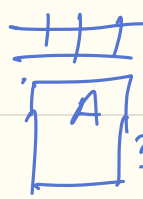
② $m > n$ $y_1 = a_{11}x_1 + \dots + a_{1n}x_n$

$$\vdots$$

$$y_m = a_{m1}x_1 + \dots + a_{mn}x_n$$

$m = 3 \quad n = 2$

$x_1 = \checkmark$
 $x_2 = \checkmark$

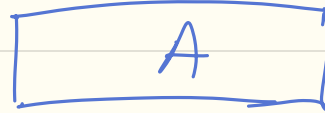
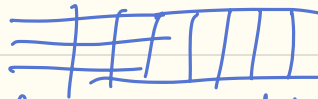


3x2 - Tall

③

$m < n$,

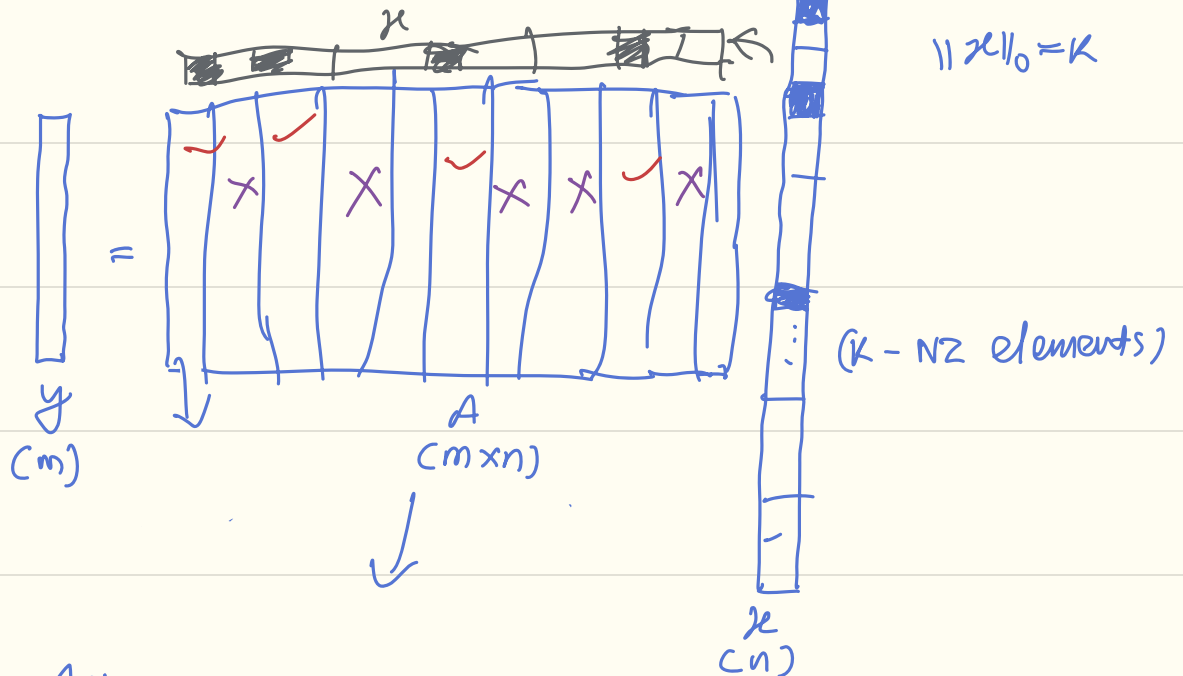
$(n-m)$ free variables.



Flat.

$\Rightarrow$

$m < n. \quad \|x\|_0 \leq k. \text{ (Sparsity)}$



$$y = Ax$$

$$\Rightarrow y = \sum_{i=1}^n x_i A_i$$

$A_i \rightarrow \underline{\text{column}}$

$A, y \rightarrow x?$

$$y = Ax \Rightarrow \hat{x} = A^T y$$

$$\Rightarrow P0: \min_x \|x\|_0 \quad \text{s.t.} \quad \|y - Ax\|_2 \leq \epsilon$$

Tolerance

NP-hard problem.

(APPROX)

$$\Rightarrow P1: \min_x \|x\|_1 \quad \text{s.t.} \quad \|y - Ax\|_2 \leq \epsilon$$

$$J(x) = \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1$$

$\lambda \rightarrow$ Regularizing Parameter.

Data fidelity

Sparse Regularized.

$g(x)$

$h(x)$

$$J(x) = g(x) + h(x)$$

$$h(x) = \lambda \|x\|_1$$

$$\partial_x h(x) = \begin{cases} 1 & x > 0 \\ [-1, 1] & x = 0 \\ -1 & x < 0 \end{cases}$$

(Convex)

(Not smooth)

$$g(x) \rightarrow \beta\text{-smooth.}$$

$$\partial_x g(x) \rightarrow \text{has Lipschitz } \beta.$$

$$\beta \geq \max(\text{eigen}(A^T A))$$

(convex)

(smooth)

$$J(x) \rightarrow \text{convex} + \text{not smooth}$$

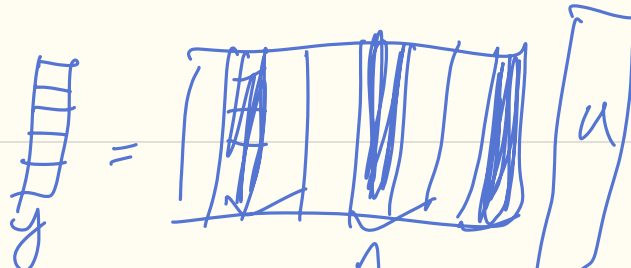
$$\min_x J(x) = \min_x \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1$$

① Sub-Gradient

$$x^{t+1} = x^t - \eta \underline{g'}$$

$$g' \in \partial_x g(x)$$

② Proximal $\rightarrow$ ISTA $\rightarrow$ LISTA.





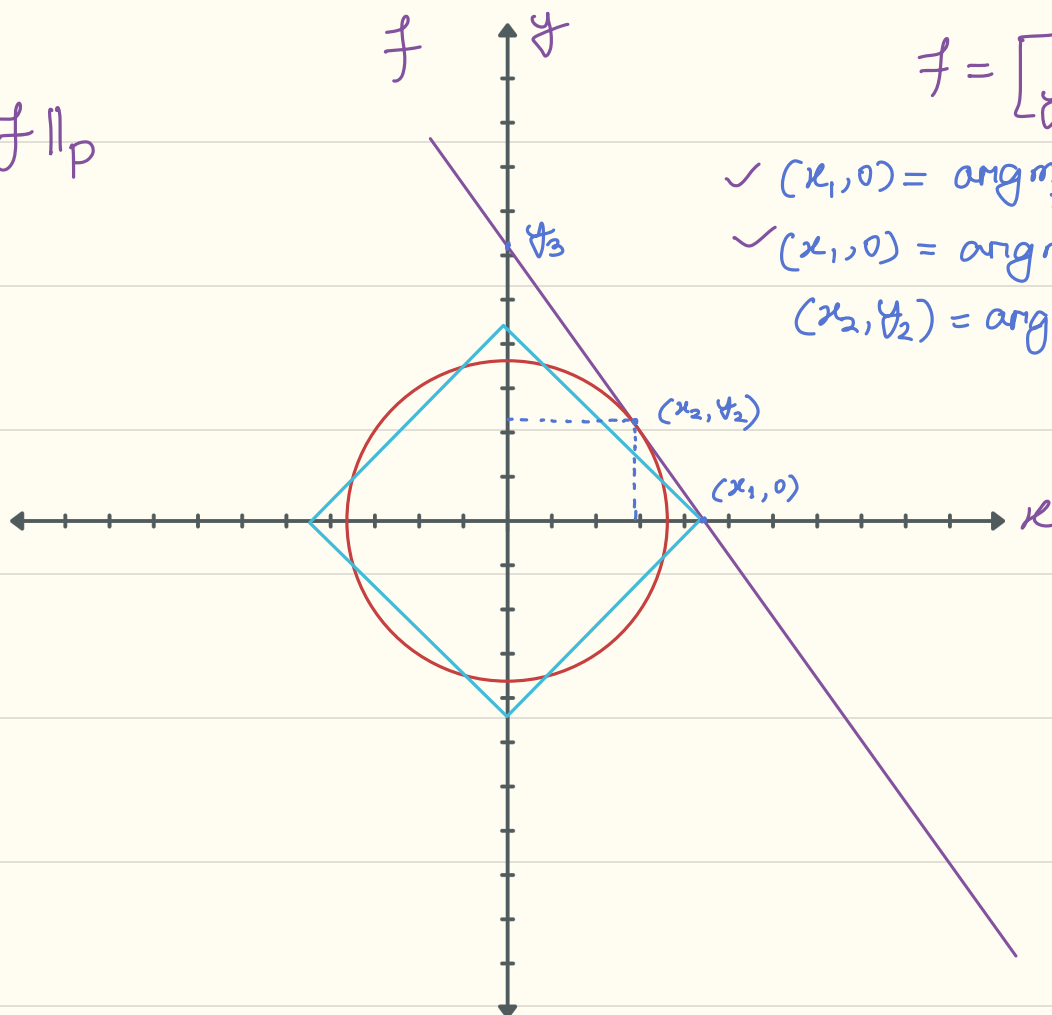
$$f = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\checkmark (x_1, 0) = \arg \min_z \|z\|_0 \quad \parallel \text{ sparse}$$

$$\checkmark (x_1, 0) = \arg \min_f \|f\|_1$$

$$(x_2, y_2) = \arg \min_f \|f\|_2$$

↳ NOT sparse.



① $y = Ax$

$$J(u) = \frac{1}{2} \|y - Au\|_2^2 + \lambda \|u\|_1$$

$$\begin{array}{c} \overline{f(n)} \\ \downarrow \\ 1 \end{array} \quad \begin{array}{c} \overline{h(n)} \\ \downarrow \\ \mathcal{R}(n) \end{array}$$

$$x^t \xrightarrow{\downarrow} x^{t+1} \rightarrow \text{cloud}$$

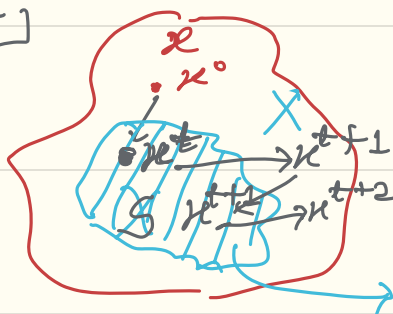
$$v \in \mathbb{R}^n \mapsto \lambda \|u\|_2$$

$$x^t \rightarrow [0, 1.9]$$

2

$$x^t \rightarrow [0, 1]$$

$$S \subseteq X$$



$$\min \|u\|_2$$

$$n+1$$

$$x^{t+1} = \text{Prox}_{\lambda h(\cdot)} \left(x^t - \eta \nabla g(x^t) \right)$$

$$= \text{Prox}_{\eta \lambda \| \cdot \|_1} \left(x^t - \eta (-A^T(y - Ax)) \right)$$

$$= \text{Prox}_{\eta \lambda \| \cdot \|_1} \left(x^t - \eta A^T A x + \eta A^T y \right)$$

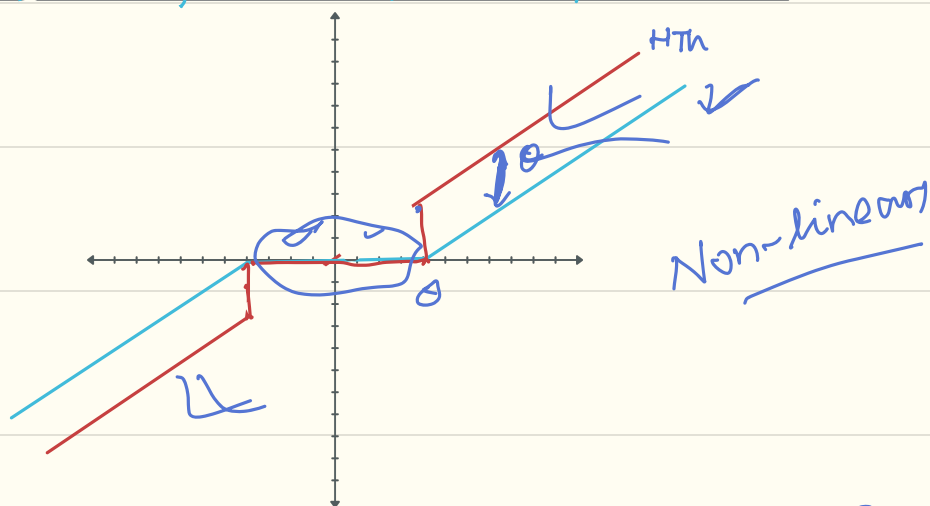
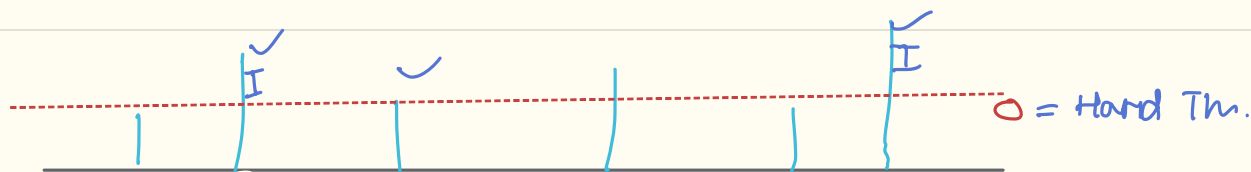
$$= \text{Prox}_{\eta \lambda \| \cdot \|_1} \left((I - \eta A^T A) x^t + \eta A^T y \right)$$

$$\eta = \frac{1}{\beta}$$

$$x^{t+1} = \mathcal{S}_{\eta \lambda} \left[(I - \eta A^T A) x^t + \eta A^T y \right]$$

$$\beta \geq \max(\text{eigen}(A^T A))$$

$$\mathcal{S}_{\theta}(x_i) = \text{sign}(x_i) \max(|x_i| - \theta, 0)$$



(PF) →

$$\min_x J(x) \Rightarrow \hat{x} = \arg \min \|y - Ax\|_2^2 + \lambda \|x\|_1$$

ISTA

$(A, y), \lambda$

$$x^{t+1} = \mathcal{S}_{\lambda} \left((I - \eta A^T A) x^t + \eta A^T y \right)$$

$$\mathcal{S}_{\lambda}(x_i) = \text{sign}(x_i) \max(|x_i| - \lambda, 0)$$

Fooswar model

$$y = Ax, \quad x \in \mathbb{C}^n(\mathbb{R}^n), \quad A \in \mathbb{R}^{m \times n}$$

$$y \in \mathbb{C}^m(\mathbb{R}^m), \quad \|x\|_0 \leq K.$$

ISTA $\rightarrow$ Initializ. $\rightarrow x^0 = 0 \quad y = y \quad$ (Fast-ISTA)

$$x^1 = S_\lambda(\eta A^T y)$$

$$x^2 = S_\lambda((I - \eta A^T A)x^1 + \eta A^T y)$$

$$\| \Delta x \|_2 = \| x^{t+1} - x^t \|_2 \leq \epsilon \rightarrow 10^{-5}$$

SYN:

$$m = 30$$

$$n = 100$$

$$K = 3/5$$

$$m = K \log(n)$$

Ideally

$$m = 5 \log 100$$

$$A \rightarrow 30 \times 100$$

$$= 10$$

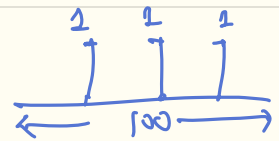
$$\text{Dataset} \rightarrow \mathcal{D}: (x_i, y_i)_{i=1}^N \triangleq \{ y_i = Ax_i + \epsilon, \|x_i\|_0 \leq K \}$$

① $A \rightarrow$ Random $\rightarrow$ column of $A \rightarrow n$
($m \times 1$)

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

($\epsilon \in \mathbb{R}$)

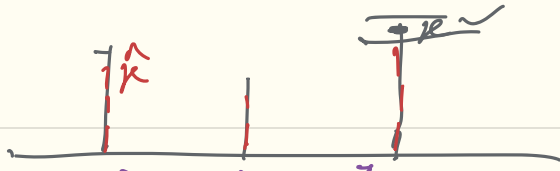
② $x \rightarrow K=3, n=100, \text{Amp } A=1.$
 $\mathcal{U}[0.7, 1.3]$



$$\textcircled{3} \quad y = Ax$$

(N)

Stem plot



$$\text{NMSE} = \frac{\|x^* - \hat{x}\|_2}{\|x^*\|_2}$$

(dB)
($20 \log \text{NMSE}$)

$$\text{HR} = \frac{|\text{Index NZ}(x^*)|}{|\text{Index NZ}(\hat{x})|}$$

(%)

$$I \rightarrow \|x_I - \hat{x}_I\| \leq 0.1$$

* LISTA

$$x^{t+1} = S_\lambda((I - \eta A^T A)x^t + \eta A^T y)$$

$\downarrow$
NL
Scalar

$$W_2$$

($n \times n$)

$$W_1$$

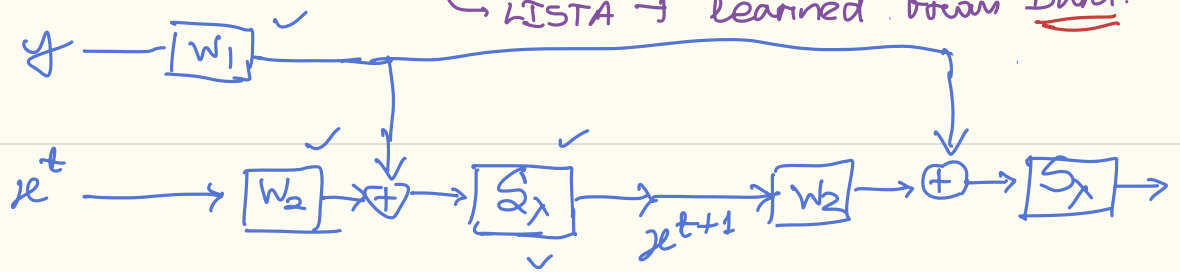
($n \times m$)

$$x^{t+1} = S_\lambda(W_1 y + W_2 x^t)$$

(X)

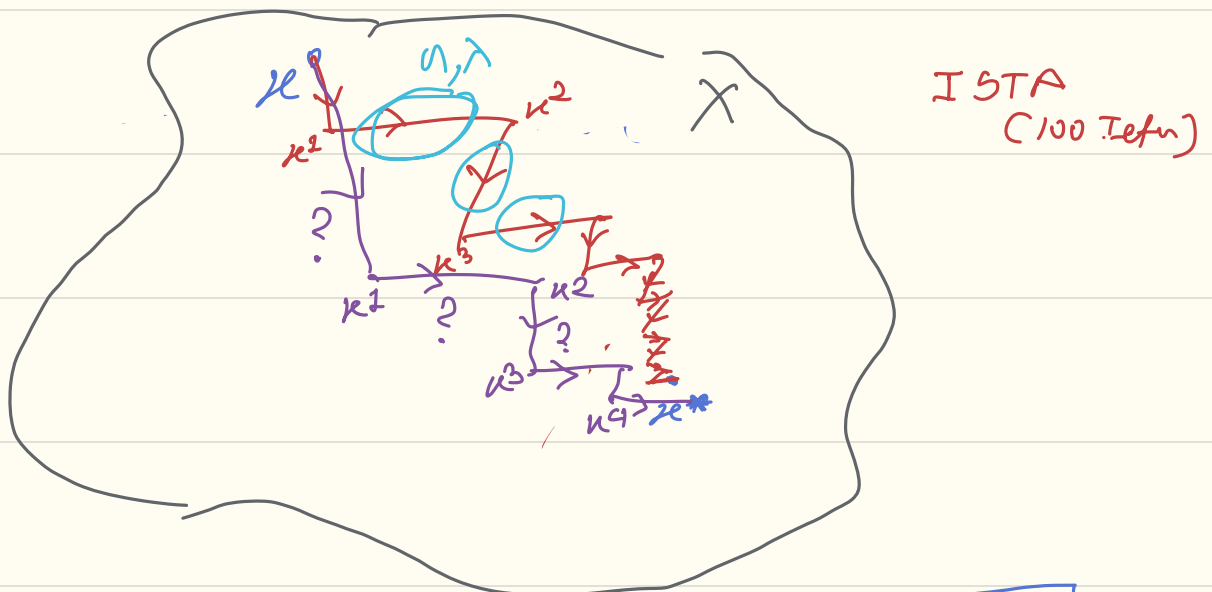
$\rightarrow$ ISTA $\rightarrow$ Empirical

Don't

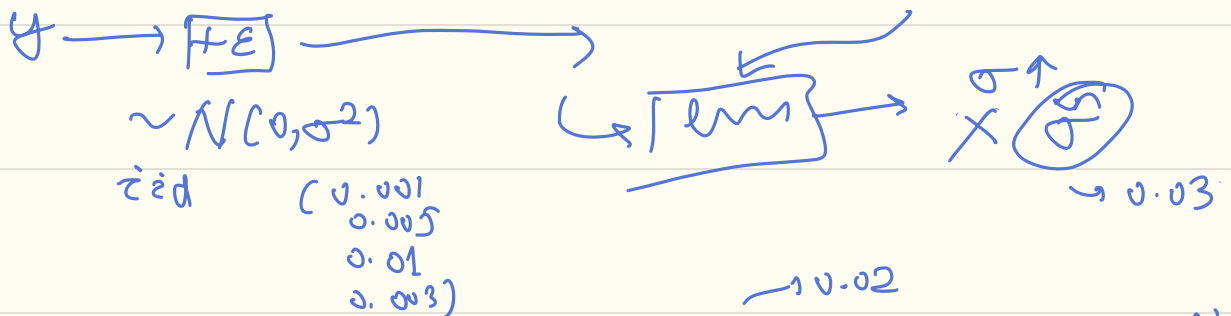


$$\min \frac{1}{N} \sum_{i=1}^N \|\hat{x} - x^*\|_2^2 \rightarrow \text{MSE loss.}$$

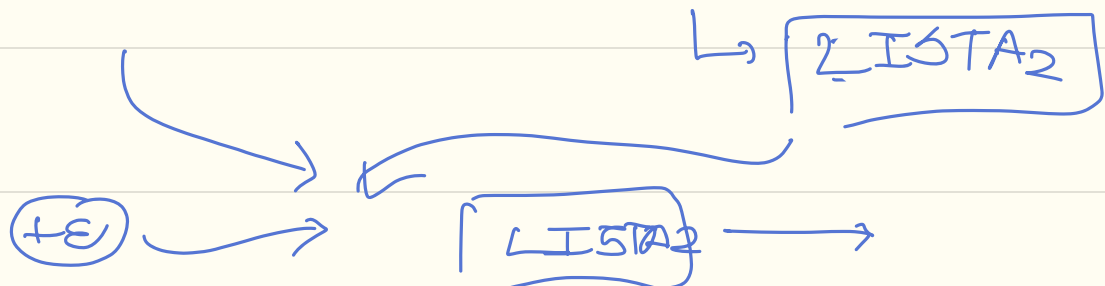
$T \rightarrow \text{layers} \rightarrow 10-15$ (Ex - 10 layers LISTA)



$$y = Ax \rightarrow \mathcal{D}_1 = \{x_i, y_i\}_{i=1}^N \rightarrow \boxed{\text{LISTA}_1}$$



$$y = Ax + E, \quad E \sim N(0, \sigma^2) \rightarrow \mathcal{D}_2 = \{ \}_{i=1}^N$$



* ISTA , ADMM , Lagrangian ... , Primal and dual ,
HQS ,

* Unfolded $\rightarrow$ Vishal Monga
()